

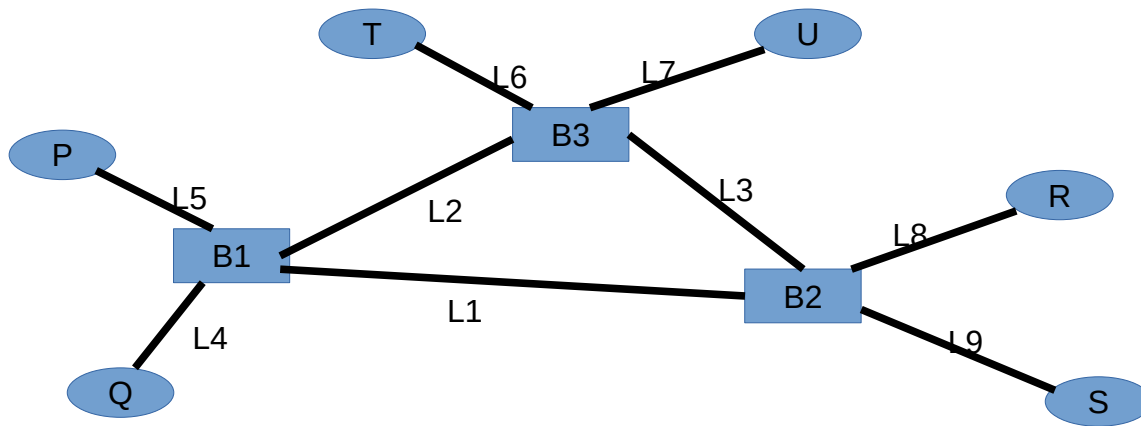
CS348 Computer Networks Quiz-2, Thu 10 Sep 2026, 09.30am-10.10am, Max. Marks: 10

Name: _____ Roll number: _____ House: _____

General instructions

- Write only in the space provided. Answer briefly but crisply. Write neatly and clearly.
- You are allowed to refer to your own hand-written notes only. You do NOT need a calculator.
- All answers have to be (briefly) explained. State any necessary assumptions.
- The final answer of each question has to be entered in the given box.

Q1: [1 x 4 = 4 marks] Learning bridges and spanning tree: In the ethernet network below, B1, B2, B3 are bridges and other nodes P, Q, R, S, T, U are end-hosts. The various links are labeled as shown. In the questions below, assume: (a) spanning tree protocol has stabilized, (b) forwarding tables are empty to begin with.



1. Which port of which bridge is inactive (if any)?

Answer:

Reason:

2. Node P transmits a frame destined to node S. How many links is the frame sent on (assume no losses)?

Answer:

Reason:

3. After above, S transmits a frame back to P. How many links is this frame sent on (assume no losses)?

Answer:

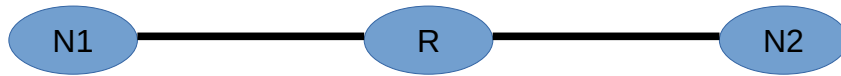
Reason:

4. After above, T transmits a frame to S. How many links is this frame sent on (assume no losses)?

Answer:

Reason:

Q2: IP fragmentation and pipelining [1+1+2+1+1=6 marks]: In the network below, N1 sends a packet P1 with a 620 byte TCP payload to N2 via IP router R. The two links N1-R and R-N2 are both of the same technology, with a link capacity of 4Mbps. Assume that TCP and IP headers are both 20 bytes each. Ignore propagation and processing delays, and assume that there is no other traffic in the network.



1. What is the minimum required MTU of the link technology so that P1 does not get fragmented by IP ?

Answer:

Reason:

2. What is the delay (in ms) experienced by P1, from N1 to N2, in the above no-fragmentation scenario?

Answer:

Calculation:

3. For what value of MTU of the link technology does P1 get fragmented into two IP fragments of same size?

Answer:

Reason:

4. What is the delay (in ms) experienced by P1, from N1 to N2, in the above two-fragments scenario?

Answer:

Calculation:

5. In which of the two scenarios (no-fragmentation vs two-fragments) is P1's delay lower? Explain the intuition behind the same.

Answer:

Explanation:
